Netflix Content Analysis: EDA & Business Insight using Python

1. 🔍 Problem Statement

With the rapid growth of streaming platforms like Netflix, understanding content trends, user preferences, and production strategies has become crucial for data-driven decision making.

However, raw datasets alone cannot provide actionable insights without proper analysis and visualization.

2. 🎯 Objective

This project aims to perform Exploratory Data Analysis (EDA) on Netflix content data to uncover:

* Content distribution (Movies vs TV Shows)

* Growth trends over time

* Top contributing countries

* Popular genres and ratings

* Business insights for content strategy

3. Import Required Libraries

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


sns.set(style="whitegrid")
```

4. Upload Dataset

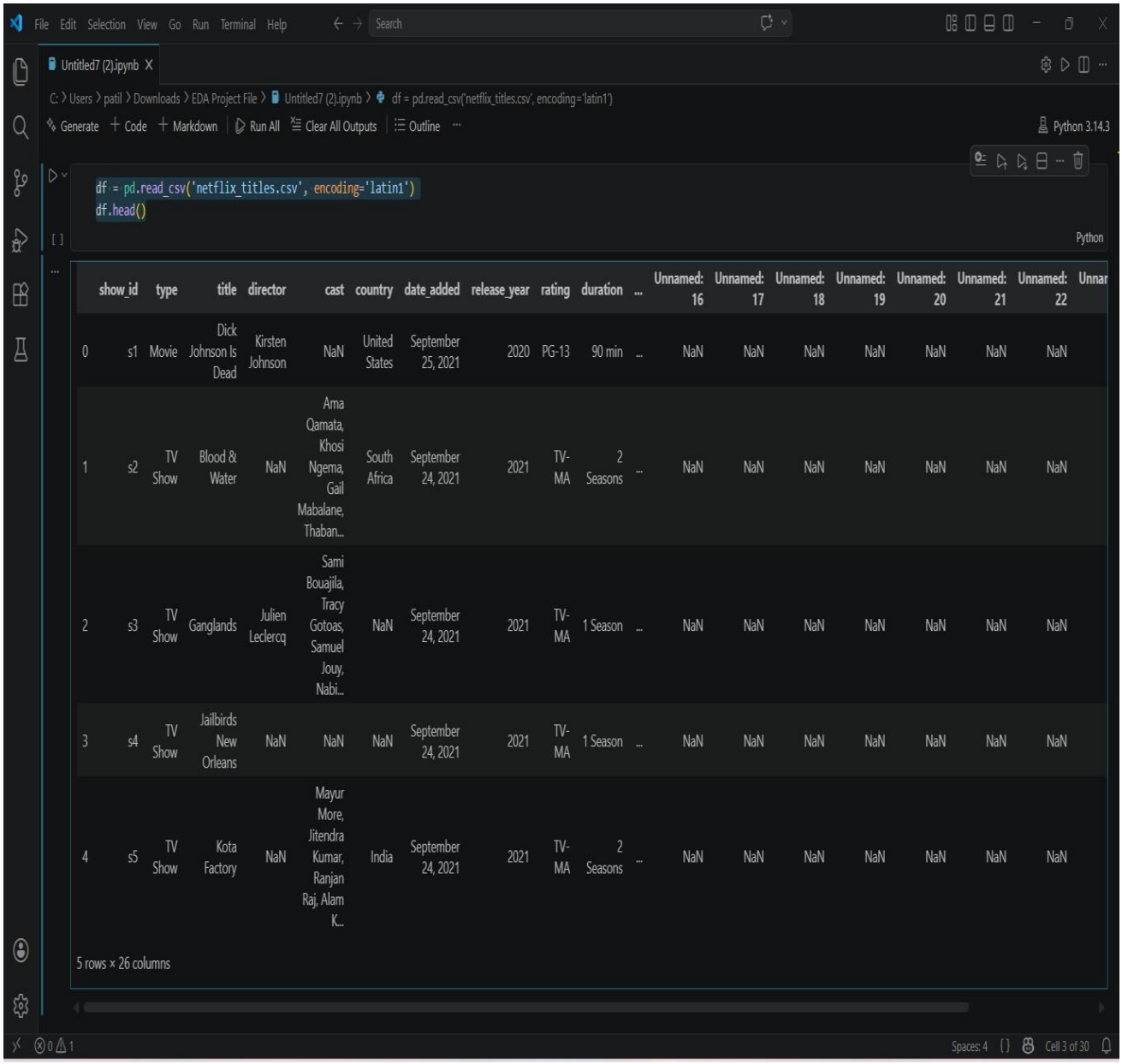
```
from google.colab import files

uploaded = files.upload()
```

5. Loading the Dataset

df = pd.read_csv('netflix_titles.csv', encoding='latin1')

df.head()



	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	...	Unnamed: 16	Unnamed: 17	Unnamed: 18	Unnamed: 19	Unnamed: 20	Unnamed: 21	Unnamed: 22	Unnar
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

6. Data Overview & Missing Value Analysis

<class 'pandas.core.frame.DataFrame'>

📌 Dataset Overview:

* Total Records: 8800+

* Features: 12 useful columns after cleaning

* Data Types: Categorical + Numerical

⚠️ Missing Values:

* Director: 30%

* Country: ~10%

* Cast: ~9%

🛠 Cleaning Steps:

* Removed unnecessary "Unnamed" columns

* Filled missing categorical values with placeholders

* Converted date_added to datetime format

7. Data Cleaning and Dataset Preparation

df = df.loc[:, ~df.columns.str.contains('^Unnamed')]

df.head()

[ ] Python

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

☐ Data Cleaning Process:

1. Removed irrelevant columns (Unnamed columns)

2. Handled missing values:

   • Director → "Unknown"

   • Cast → "Not Available"

   • Country → "Unknown"

3. Converted date_added into datetime format

4. Created new feature: year_added

💡 Why this matters:

Clean data ensures accurate analysis and prevents misleading insights.

8. Handling Missing Values (Data Imputation)

df['director'].fillna('Unknown', inplace=True)

df['cast'].fillna('Not Available', inplace=True)

df['country'].fillna('Unknown', inplace=True)

df['rating'].fillna('Not Rated', inplace=True)

(Missing values in important categorical columns were handled using appropriate placeholder values to maintain data consistency and avoid loss of information)

9. Date Feature Engineering & Data Type Conversion

df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')

df.head()

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Not Available	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
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4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

10. Content Distribution Analysis (Movies vs TV Shows)

df['type'].value_counts()

type	count
Movie	6132
TV Show	2677

dtype: int64

11. Bar Chart: Distribution of Movies and TV Shows on Netflix

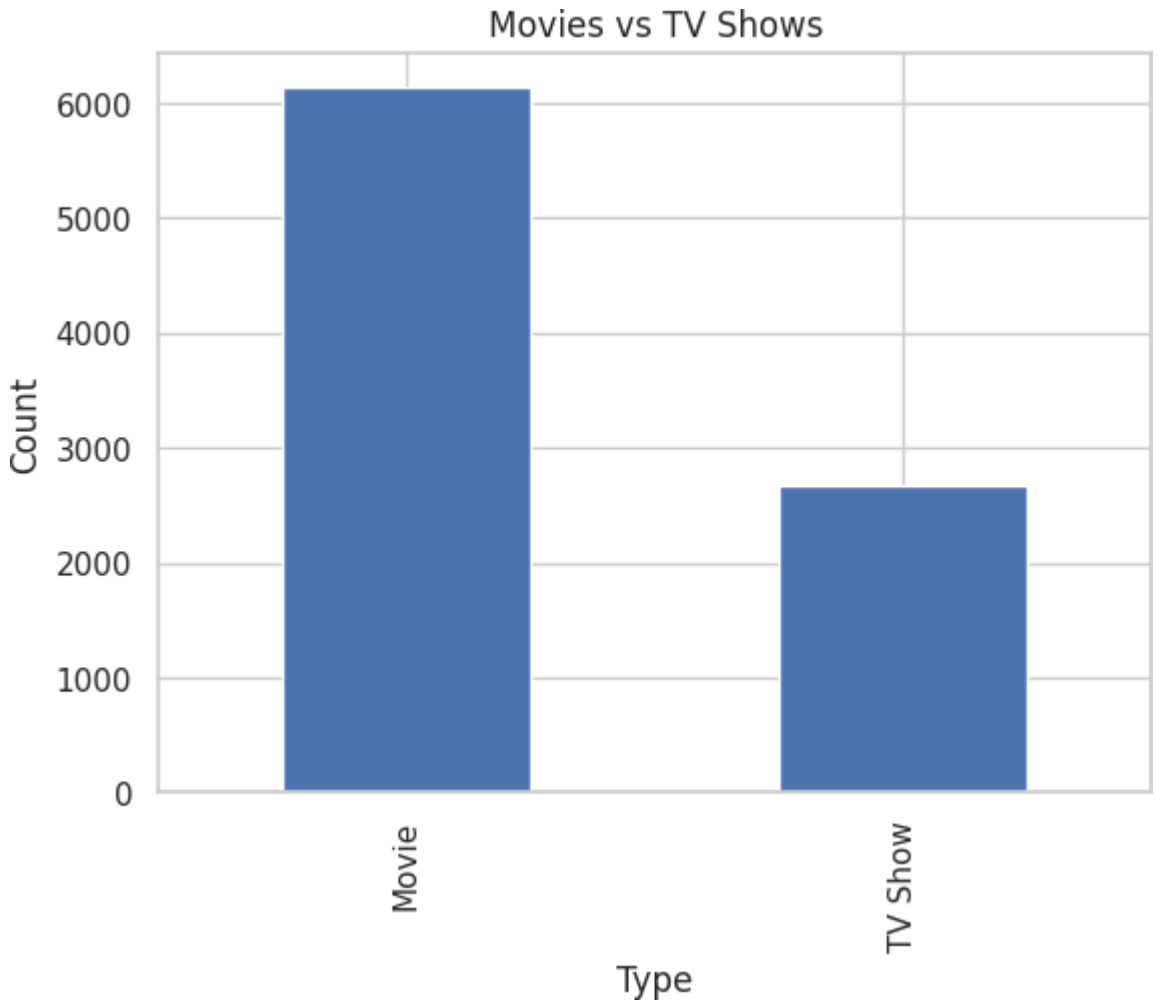
```
df['type'].value_counts().plot(kind='bar')

plt.title('Movies vs TV Shows')

plt.xlabel('Type')

plt.ylabel('Count')

plt.show()
```



📊 Insight:

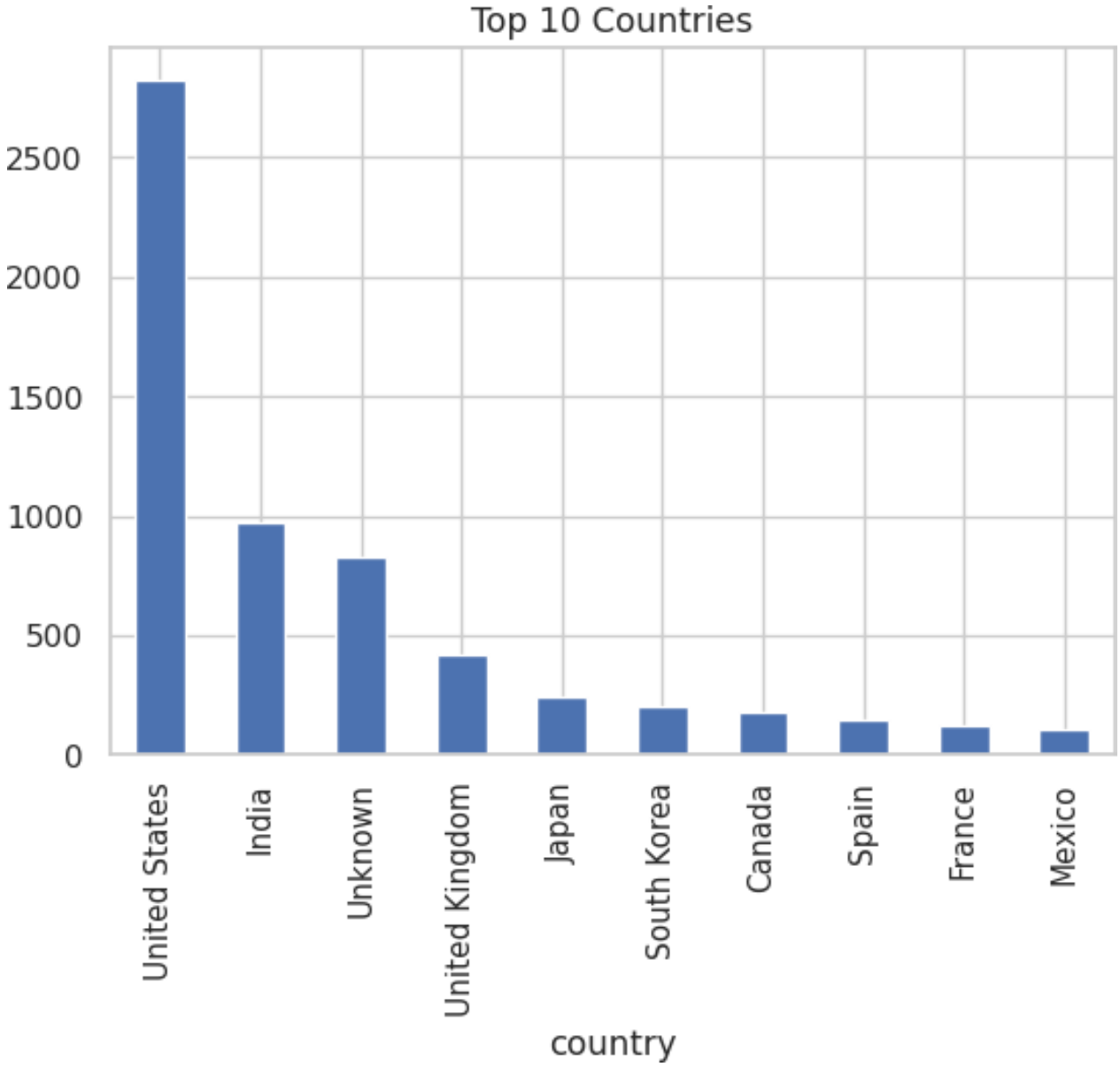
Netflix has significantly more Movies (~70%) than TV Shows.

📌 Business Impact:

This indicates Netflix focuses more on short-term, high-volume content production rather than long-term TV series investments.

12. Top 10 Countries by Number of Netflix Titles

```
df['country'].value_counts().head(10).plot(kind='bar')
plt.title('Top 10 Countries')
plt.show()
```



🌍 Insight:

The United States dominates Netflix content production, followed by India and the UK

📌 Business Impact:

Netflix heavily relies on Western markets but is expanding in emerging markets like India.

13. Content Distribution by Release Year (Last 10 Years)

df['release_year'].value_counts().sort_index().tail(10)

release_year	count
2013	288
2014	353
2015	560
2016	902
2017	1032
2018	1147
2019	1030
2020	953
2021	592
2024	1

dtype: int64

dtype: int64

📈 Insight:

Content production increased rapidly after 2015 and peaked around 2019-2020.

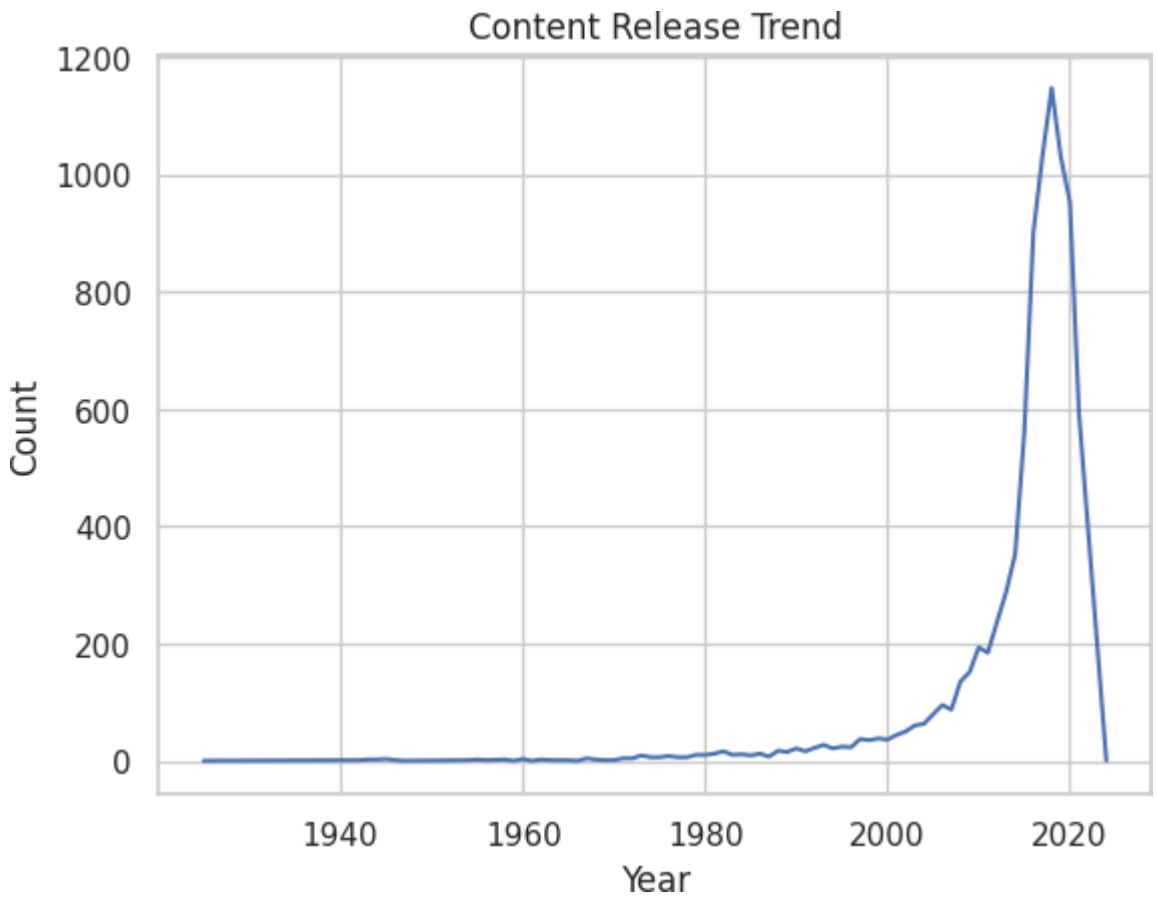
📌 Business Impact:

This reflects Netflix's global expansion phase and aggressive content strategy.

14. Line Chart: Trend of Content Releases Over Time

df['release_year'].value_counts().sort_index().plot()

```
plt.title('Content Release Trend')
plt.xlabel('Year')
plt.ylabel('Count')
plt.show()
```



Growth in Content Production

- If the line is increasing over time, it indicates:
  - The platform (or dataset source) is expanding its content library
  - More investment in content creation/acquisition in recent years
    ✅ Business meaning: Growth strategy is working; content catalog is expanding to attract users.

✅ **Business meaning: Helps identify successful periods of high content investment.**

- Years with spikes show maximum content releases
- These peaks often indicate:

- Aggressive content acquisition
- Strategic expansion phases

✅ Business meaning: Helps identify successful periods of high content investment.

15. Top 10 Content Ratings by Number of Titles

```
df['rating'].value_counts().head(10)
```

rating	count
TV-MA	3208
TV-14	2160
TV-PG	863
R	799
PG-13	490
TV-Y7	334
TV-Y	307
PG	287
TV-G	220
NR	80

dtype: int64

🎬 Insight:

TV-MA and TV-14 are the most common ratings.

📌 Business Impact:

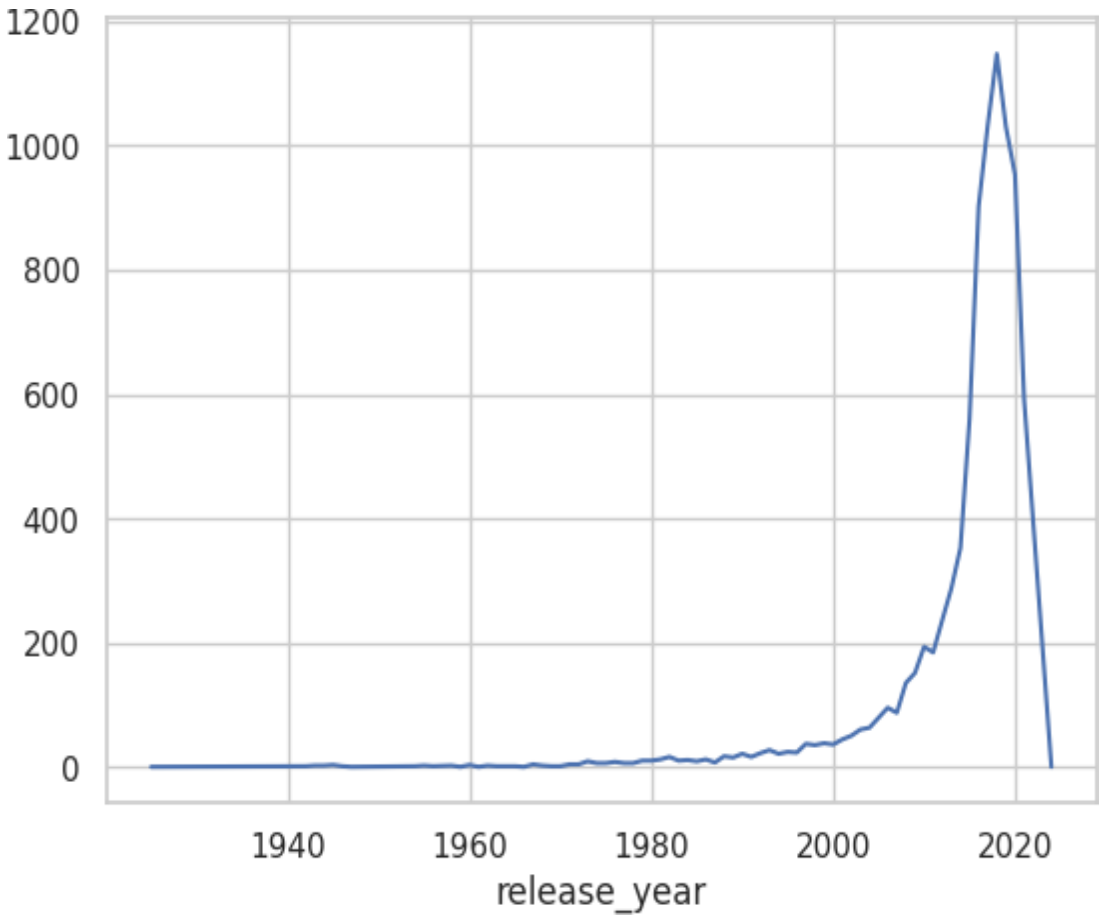
Netflix primarily targets mature audiences, indicating a focus on adult-oriented content.

16. Exporting Cleaned Dataset to CSV File

```
df.to_csv('cleaned_netflix_data.csv',  index=False)
```

17. Line Chart: Year-wise Distribution of Netflix Content

```
df.groupby('release_year')['type'].count().plot()
```



18. Country-wise Content Distribution by Type (Movies vs TV Shows)

Content type by country

df.groupby(['country','type']).size().unstack().head()

type	Movie	TV Show
country		
, France, Algeria	1.0	NaN
, South Korea	NaN	1.0
Argentina	58.0	18.0
Argentina, Brazil, France, Poland, Germany, Denmark	1.0	NaN
Argentina, Chile	2.0	NaN

print("Insight: Netflix has significantly more Movies than TV Shows, indicating higher demand for movie content

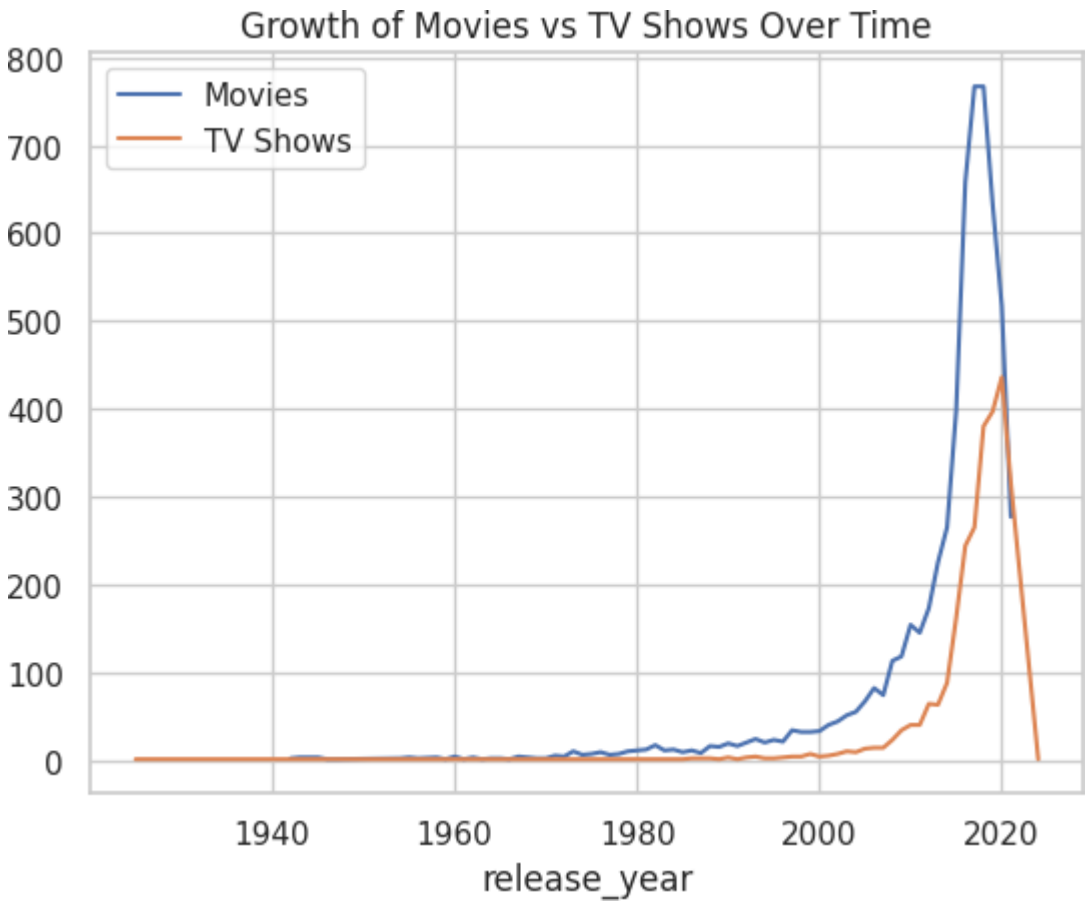
19. Line Chart: Comparative Growth of Movies and TV Shows Over Time

movies = df[df['type'] == 'Movie']

tv = df[df['type'] == 'TV Show']

```
movies.groupby('release_year').size().plot(label='Movies')
tv.groupby('release_year').size().plot(label='TV Shows')

plt.legend()
plt.title("Growth of Movies vs TV Shows Over Time")
plt.show()
```



📊 Insight:

Both Movies and TV Shows increased over time, but Movies grew faster.

📌 Business Impact:

Movies are easier to scale compared to TV Shows, making them a preferred content type.

20. Bar Chart: Top 10 Most Popular Genres on Netflix

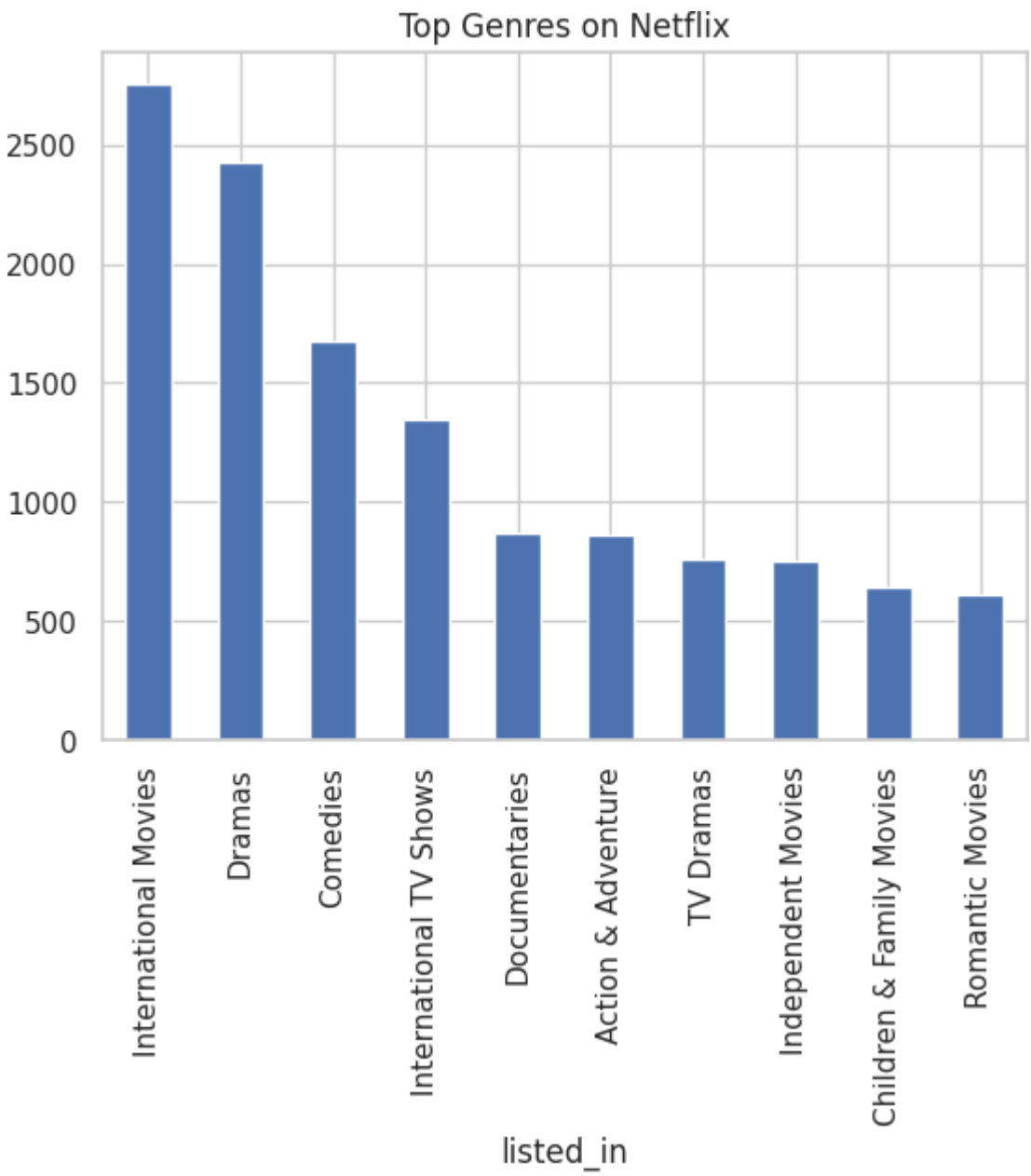
```
from collections import Counter

genres = df['listed_in'].str.split(', ').explode()
top_genres = genres.value_counts().head(10)

top_genres.plot(kind='bar')
```

```
plt.title("Top Genres on Netflix")

plt.show()
```



🎭 Insight:

International Movies, Dramas, and Comedies are the most popular genres.

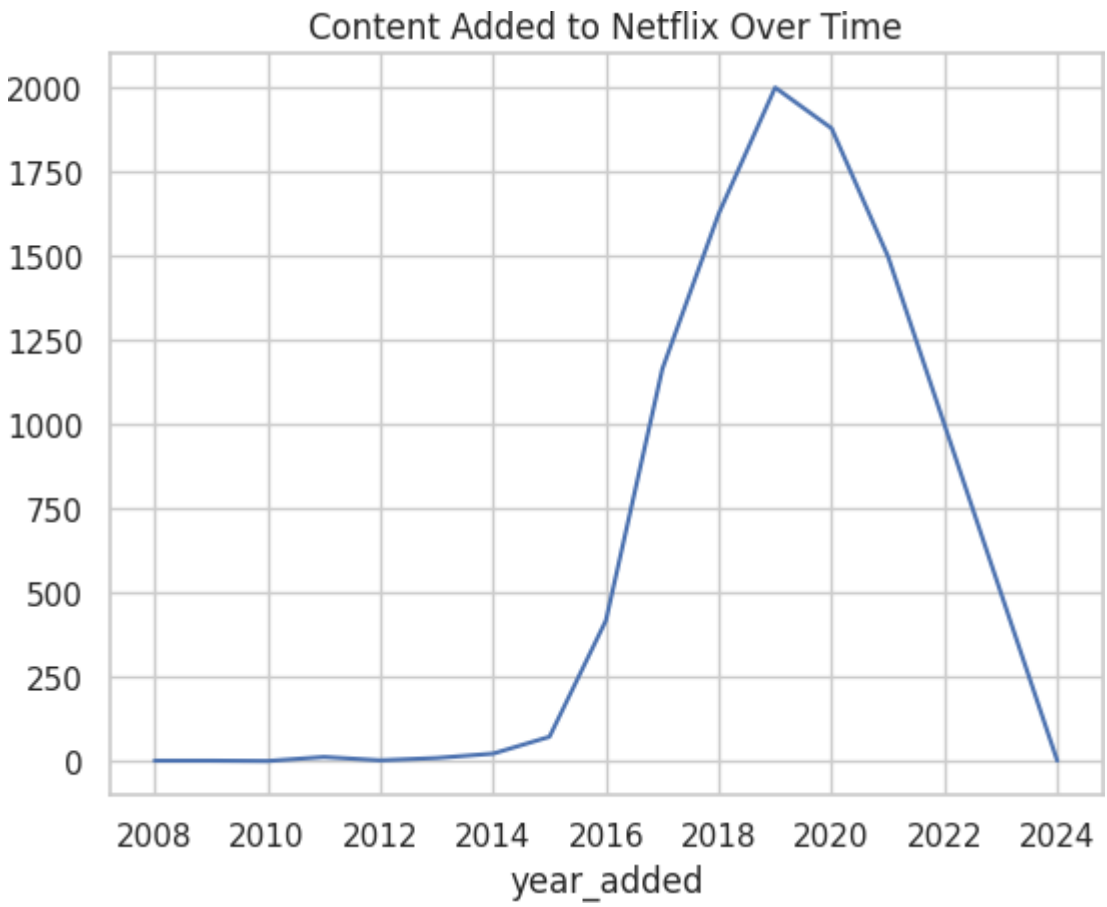
📌 Business Impact:

Netflix prioritizes globally appealing genres to attract a diverse audience.

21. Line Chart: Trend of Content Added to Netflix Over the Years

```
df['year_added'] = df['date_added'].dt.year
```

```
df['year_added'].value_counts().sort_index().plot()
plt.title("Content Added to Netflix Over Time")
plt.show()
```



📈 Insight:

Content additions peaked around 2019–2020 and declined afterward.

📌 Business Impact:

This may reflect market saturation or production slowdowns post-pandemic.

22. Display the Visualization Output

```
# Displaying the generated plot

plt.show()

print("""
Business Insights:
```

1. Netflix focuses heavily on Movies → Content strategy insight

2. USA dominates → Market dependency

3. Growth after 2016 → Platform expansion phase

4. Popular genres → Helps recommendation systems

""")

23. Summary of Key Business Insights

🚀 Key Business Insights:

1. Netflix focuses heavily on Movies (~70%), indicating a scalable content strategy.

2. The USA dominates production, but India is emerging as a key market.

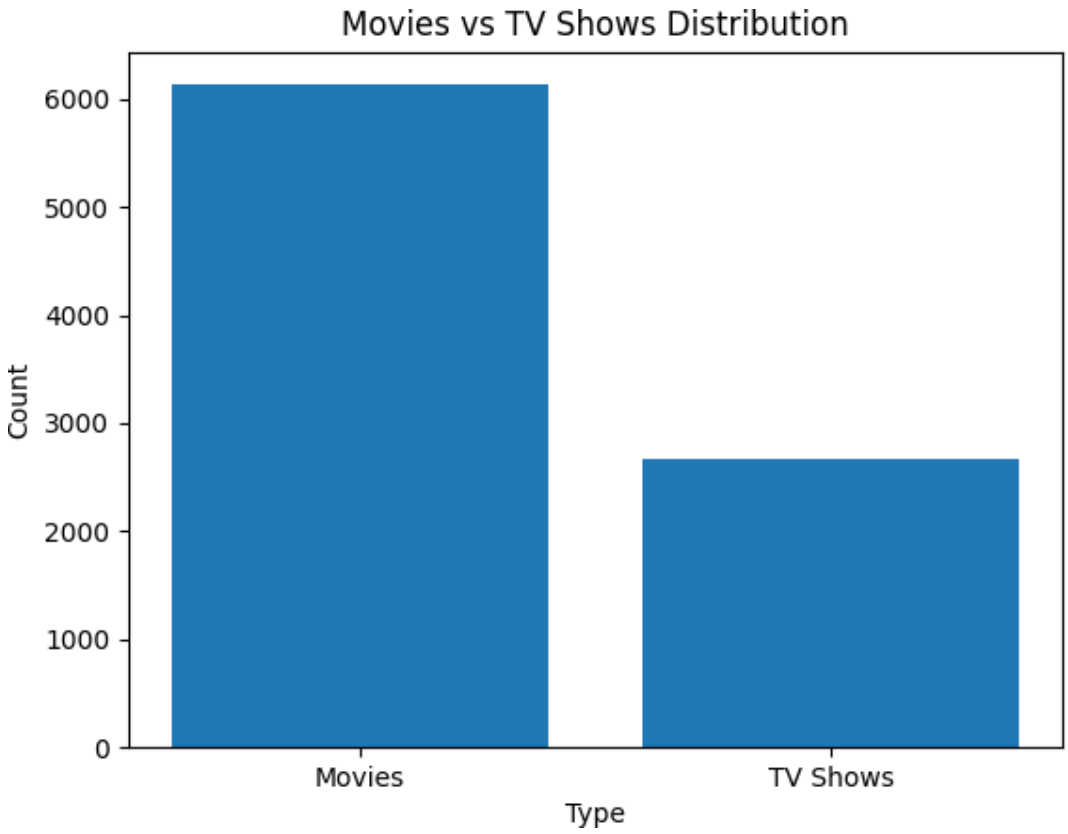
3. Content growth surged after 2015, showing aggressive expansion.

4. Mature audience content (TV-MA, TV-14) dominates the platform.

5. International genres are gaining popularity, reflecting global audience targeting.

24. Final Comparison Visualization

plt.savefig("movies_vs_tv.png")



25. Top 10 Most Featured Actors on Netflix

```
actors = df['cast'].str.split(', ').explode()
```

actors.value_counts().head(10)

cast	count
Not Available	825
Anupam Kher	43
Shah Rukh Khan	35
Julie Tejwani	33
Naseeruddin Shah	32
Takahiro Sakurai	32
Rupa Bhimani	31
Akshay Kumar	30
Om Puri	30
Yuki Kaji	29

dtype: int64

🎭 Insight:

Certain actors appear frequently, indicating repeated collaborations.

📌 Business Impact:

Frequent actor appearances may influence viewer engagement and brand consistency.

26. 📌 Conclusion

This analysis highlights Netflix's strategic focus on scalable movie content, global expansion, and targeting mature audiences.

The platform's growth trend indicates strong investment in content between 2015-2020, followed by stabilization.

27. 🧠 Future Improvements

- Build a recommendation system
- Perform sentiment analysis on descriptions
- Create an interactive dashboard (Power BI / Tableau)
- Predict content success using ML .